

ChangeLoop

In a zero-discharge dyeing factory, the coal bill is set by the salt that goes in, not the water that comes out. This is a tool that shows the planner that cost while they are still choosing.

1. Title page

Startup name	ChangeLoop
Tagline	In a closed loop, salt sets the coal bill.
Date	9 October 2026
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Working prototype	https://changeloop-water-intelligence.onrender.com

2. Background and problem statement

Context

Tirupur in Tamil Nadu is India's knitwear capital. In the year to March 2026 it exported around Rs 46,000 crore of knitwear, more than half of everything India sells abroad in cotton knitwear. Its dyeing factories sit on the Noyyal river.

In 2011 the Madras High Court shut around 700 dyeing and bleaching units for putting salty waste water into the river. They were allowed back only if they stopped releasing liquid waste completely. The rule is called zero liquid discharge, or ZLD.

Today about 300 dyeing units run on 18 shared treatment plants, and another 60 have their own. Between them they recycle about 120 million litres of water a day and recover around 94% of it. Reported unit counts vary between sources, from roughly 360 to 450, so I treat the cluster as a few hundred units rather than a precise number.

It worked for the river. The waste water stopped going in. But the problem moved to the boiler house, and nobody is managing it.

The problem

To recycle the water, a factory has to boil the leftover salty stream until only dry salt is left. Boiling needs heat, the heat comes from steam, and in Tirupur the steam mostly comes from coal. Boiling one cubic metre of leftover takes about 188 kWh of heat.

Here is the part almost nobody acts on.

The amount you have to boil is fixed by the salt, not by the water.

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leftover volume = salt mass / strongest the membranes can make it
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Water goes round the loop and comes back. Salt does not. Salt has to leave as a solid, and the only way out is through the evaporator. So the evaporator must boil whatever amount of water that salt happens to be sitting in.

The result is easy to test, and my system tests it on every request:

Change a factory makes	What happens to evaporator energy
Uses 20% less water	no change at all
Puts in 20% less salt	20% less energy

Zero, not a small amount. It falls straight out of the arithmetic.

So two changes that look identical on a water meter do completely different things to the coal bill. Most water-saving work in the cluster aims at the water. It does not touch the coal.

Why it has not been fixed

The salt load is decided by a production planner, choosing which dye lots run in what order and which rinse method to use. They make that call hours before anything shows up anywhere.

The coal bill arrives weeks later, in a different department's budget, with no way to trace it back to the choice that caused it. The planner is not being careless. They are working without the one number that would change their mind.

There is a second problem one level up. A factory runs several dyeing machines into one shared evaporator. Each planner picks what is best for their own machine, and every choice is correct on its own. Added together they can still be more than the evaporator can boil. In my model four machines each picking their own best plan put the shared evaporator at **121.9% of what it can handle**. No tool that looks at one machine can see this.

Target audience

The people most affected are the ones running these factories under a rule they did not choose and cannot opt out of.

Who	Why this matters to them
Production planners in Tirupur dyeing units	They make this decision every shift and cannot see what it costs
Factory owners	They pay a coal bill they cannot trace or control
The 18 shared treatment plants	They buy the steam and store the salt. Somewhere between 60,000 and 73,000 tonnes of waste salt has built up across them, with no settled disposal route
Households and farmers in the Noyyal basin	Groundwater here is heavily stressed. Every litre not pulled out matters more than a litre saved somewhere easy

3. Vision and mission

Vision

Every closed water loop in Indian industry is run on what it actually costs, not on what is easy to measure.

Water meters are everywhere and they measure the wrong thing. In a closed loop the environmental and financial damage is driven by dissolved load, and almost nothing in industry is set up to see that. The long-term aim is to make the hidden number visible wherever this physics applies, starting with textiles and extending to cooling towers, boiler blowdown and other closed systems.

Mission

Put the downstream cost in front of the person making the upstream decision, at the moment they make it, with every number open to inspection.

In practice that means three things. Build the decision layer and prove the physics. Prove it on one factory with real instruments. Then reach the rest of the cluster through the shared treatment plants, whose own economics already point the same way.

4. Solution overview

What ChangeLoop is

ChangeLoop is a decision-support tool that sits where the dyehouse schedule gets made. Before a shift starts it looks at the different orders the planner could run the lots in, and the different rinse methods available. For each combination it works out the whole chain: water used, salt load, leftover volume, evaporator steam, CO₂, rupees, and whether any delivery date is missed.

Then it recommends one plan, and shows its working.

What makes it different

It optimises for the consequence, not for the throughput. Existing dyehouse software plans for machine use and delivery dates. ChangeLoop adds the downstream thermal cost of the schedule, which is the part nobody currently prices.

A late delivery is not treated as a trade-off. If a plan misses a ship date, it gets removed from the search rather than given a bad score. A factory cannot accept a schedule that saves water by shipping late, so the tool never offers one.

Every input number says where it came from. All 25 of them are tagged as published, derived, assumed or measured, with the source and the arithmetic. Anyone can open the list in the running system and read it.

It fails safe. It issues no machine settings and cannot release a dye bath. Early release needs a named person to approve it. If a sensor reading is missing or implausible, the system stops and claims zero saving rather than guessing.

It only counts savings somebody approved. If the planner turns a recommendation down, it counts as zero, even though the tool already worked out what it would have saved.

It handles the shared evaporator. It compares what each machine would pick on its own against the cheapest combination that actually fits the plant, and prices the difference.

Value proposition

For a factory owner: lower coal and water bills from a scheduling change, with no equipment to buy and no change to the existing approval process.

For a planner: the number that is missing from their screen, at the moment they need it.

For a shared treatment plant: less salt arriving, which means less steam to buy and less solid salt to store.

For a regulator: a tool that quantifies what a price change would actually do, instead of arguing for one in principle.

5. Objectives and goals

Short term, the next 12 months

Objective	Measure of success
Install instruments at one pilot factory	Steam flow and reject conductivity metered for 28 days at over 95% completeness
Prove the recommendations are sound	Planner agrees with over 60% of them unprompted, during four weeks where the tool changes nothing
Measure a real saving	Water and steam reduction measured with a confidence interval, on alternate weeks on and off
Keep quality intact	Zero colour-fastness failures attributable to an approved plan
Turn assumptions into measurements	Move the two heaviest input numbers from published to measured

Long term, two to five years

Objective	Target
Reach the cluster through shared treatment plants	40 paying units across 2 to 3 CETPs by year 3
Make the big saving the default	Low-salt chemistry adopted where the measured case supports it, worth about 47% of water and steam per shift
Solve the cluster coordination problem	A method that works for hundreds of member units, not just the 256 combinations the current solver checks
Prove the pricing argument	Published, independently reviewed evidence of what carbon or water pricing would change in this sector
Extend beyond textiles	One validated application in another closed-loop system, such as evaporative cooling

6. Business model

How it makes money

An annual software subscription per dyeing unit, sold mainly through the shared treatment plants rather than factory by factory.

The two prices below are my own proposal. They are not benchmarked against comparable software, because I could not find published pricing for anything that does this job. Everything downstream of them in this section, including the market size, moves if the price moves.

Item	Amount	Notes
Annual subscription	Rs 2,40,000 per unit	Covers the machines in scope, typically four
One-off setup	Rs 4,50,000 per unit	Connecting to the planning system and the effluent meters, plus training

Why a factory would pay that

Using today's tariffs and the plan the tool actually recommends right now, for a unit running 9,000 lots a year:

	Plan it recommends today	Low-salt plan, once the price gap closes
Gross saving a year	Rs 4,08,000	Rs 52,04,000
Less subscription	Rs 2,40,000	Rs 2,40,000
Net benefit a year	Rs 1,68,000	Rs 49,64,000
Payback on setup	2.7 years	about 5 weeks
Return on cost	37%	1,103%

I am showing both columns on purpose. The left column is what a factory gets today, and it is a modest, believable return. The right column is what the same factory gets if the low-salt dye premium falls about 12%, which is the gap my system measured. The upside is real, but it is not what I would ask anyone to sign for.

How customers are reached

Through the shared treatment plants. Tirupur has 18 of them serving about 300 units. One commercial relationship reaches roughly 17 factories, which cuts the cost of selling by an order of magnitude compared with knocking on doors.

The treatment plants also want this for their own reasons. Less salt arriving means less steam bought and less salt stored. That makes them a partner rather than just a channel.

Through running silently first. Four weeks of being right while changing nothing is how a tool earns the right to influence a factory's decisions. It costs the factory nothing.

Through trade bodies. The Tiruppur Exporters Association and the dyers' associations already coordinate on environmental compliance, which is the exact context this belongs in.

Partnerships that would help

Partner	What it brings
A shared treatment plant in Tirupur	The pilot site, the steam data, and the route to its member units
A dye chemistry supplier	Real pricing on low-salt chemistry, which is the lever with the smallest gap to close

Partner	What it brings
A state pollution control board or CPCB	Credibility, and access to the measured data that would replace my published figures
An academic water or chemical engineering group	Independent review of the model before anyone relies on it

7. Market analysis and competition

Market size

Level	Size	Basis
Tirupur, immediate	About 360 dyeing units. At my price of Rs 2,40,000 a year that is roughly Rs 8.6 crore a year	300 units on 18 CETPs plus 60 with their own plants. Other reports put the figure nearer 450
Indian textile ZLD, wider	Tirupur is one of several clusters under ZLD rules. Surat, Erode, Karur, Ludhiana, Panipat and Jetpur have similar processing bases	Directional. I have not counted units outside Tirupur and I am not going to present a number I cannot source
Adjacent closed loops, later	Evaporative cooling, boiler blowdown and mine water share the same physics	Unvalidated. Listed as direction, not as market

The growth driver is regulation that already exists. ZLD is compulsory and already paid for. ChangeLoop does not ask anyone to install new equipment, it makes equipment they already own cheaper to run.

The second driver is carbon reporting. European buyers increasingly ask Indian suppliers for emissions data. The EU carbon border mechanism does not cover textiles today, since it applies to iron and steel, cement, aluminium, fertilisers, electricity and hydrogen, but textiles have been named as a candidate for a later phase. If that happens, a tool that can show measured and traceable reductions in process emissions becomes a compliance asset as well as a cost saver. I am treating that as a possible tailwind, not as a commitment anyone has made.

Competitive landscape

Who	What they do	Why ChangeLoop is different
Dyehouse software , such as SedoMaster and Datatex	Plan production, manage recipes, track batches	They schedule for throughput and delivery. Neither prices the downstream thermal cost of the schedule, which is the whole point here
ZLD equipment companies , such as Praj, Thermax, VA Tech Wabag	Build and run the membranes and evaporators	They sell the plant. They do not touch the upstream decision that sets how hard the plant has to work
Water monitoring and IoT dashboards	Report flows and quality after the fact	They measure water volume, which the table in section 2 shows is the wrong number for the coal bill

Who	What they do	Why ChangeLoop is different
Low-salt dye suppliers , such as DyStar and Archroma	Sell the chemistry that cuts salt	They sell the lever. They cannot tell a factory when it pays, which is exactly what my system computes
Consultancies and energy audits	One-off studies	A report is read once. This runs every shift and is checked against what was actually approved

The honest summary: the pieces exist. The membranes exist, the chemistry exists, the planning software exists. What does not exist is anything that connects the upstream scheduling decision to the downstream coal bill, which is where the saving is hiding.

8. Implementation plan

Current status

The prototype is built and running. It is a live system anyone can open, not a mockup.

API	34 endpoints. Every saving is worked out on the server, so the browser cannot claim one
Tests	178, covering the physics, the safety rules, concurrency and the interface
Input numbers	25, each labelled with its source. 17 published, 4 derived, 4 assumed, 0 measured
Validation	Fed the water strength CPCB measured at a Tirupur unit, 18,340 mg/L, the model predicted 30.6% leftover. Indian plants report 20 to 30%

Timeline

Stage	Weeks	What happens	Gate to pass
1. Measure the baseline	1-4	Instruments installed. Nothing changes on the floor	28 days of data at over 95% completeness
2. Run it silently	5-8	The tool recommends, nobody acts	Planner agrees with over 60% unprompted
3. Planner may approve	9-16	Reordering only, no chemistry change	Measured water reduction, zero off-shade lots
4. Early release trial	17-24	Quality supervisor may allow early wash-off release	Every released lot lab tested. One failure ends the trial
5. Decide on chemistry	25-32	With steam cost now measured, decide if low-salt dye pays	A decision backed by measured, not assumed, numbers

Stage	Weeks	What happens	Gate to pass
6. More factories	33-52	Second and third unit on the same treatment plant	Two more units live and paying

The weeks run on and off in alternate blocks, so the comparison is against the same season, the same order mix and the same crew. The whole point is to answer one question: did the tool cause the saving, or would it have happened anyway?

Key milestones

Milestone	When
First pilot site signed	Month 1
First measured coefficient, replacing a published one	Month 2
First saving measured with a confidence interval	Month 6
First paying customer beyond the pilot	Month 10
Cluster coordination working across one CETP	Month 18
40 paying units	Month 36

9. Financial overview

All figures in this section are plans, not results. I have no revenue and no customers today.

What the first year needs

Item	Amount	What it buys
Instruments for two pilot sites	Rs 6,00,000	Steam flow meters and conductivity probes, plus installation and calibration
Site integration and engineering	Rs 5,00,000	Connecting to planning systems and effluent meters at two sites
Independent lab testing	Rs 2,00,000	Colour-fastness testing on every early-released lot
Founder stipend, 12 months	Rs 6,00,000	Full-time work on the pilot
Cloud, tooling and travel	Rs 2,00,000	Hosting, site visits, instrument servicing
Contingency	Rs 2,00,000	About 10%
Total	Rs 23,00,000	

Funding requirement

I am seeking **Rs 23,00,000** as grant or seed funding for a 12-month pilot programme across two to three dyeing units in Tirupur.

The largest single line is instrumentation, and that is deliberate. The weakness of this project today is that nothing has been measured. Two instruments at one site turn the two heaviest numbers in the model from published figures into measured ones, and nothing else buys as much credibility for as little money.

Revenue projections

These assume the pilot succeeds and the shared treatment plant channel works. Both are assumptions.

	Year 1	Year 2	Year 3
Paying units	2 at pilot rate	12	40
Subscription revenue	Rs 3,00,000	Rs 28,80,000	Rs 96,00,000
Setup revenue	included in pilot	Rs 45,00,000	Rs 1,26,00,000
Total revenue	Rs 3,00,000	Rs 73,80,000	Rs 2,22,00,000

Year 1 is a pilot year and is not meant to make money. Years 2 and 3 assume one CETP relationship in year 2 and two or three by year 3, at about 17 member units each.

10. Impact assessment

If one factory adopts it

Per shift, on the plan the tool recommends under today's prices, against a conventional single-stage rinse:

	Fresh water	Evaporator steam	Cost
Baseline	11,907 L	2,239 kWh	Rs 33,897
Recommended today	11,187 L (-6%)	2,104 kWh (-6%)	Rs 26,038
Available with low-salt chemistry	6,369 L (-47%)	1,198 kWh (-47%)	Rs 29,984

If the cluster adopts it

A straight-line projection across 400 units at 900 lots each a year. This is a projection and my system labels it as one. It assumes every factory resembles the modelled one and that every recommendation is approved. Neither has been tested, and it must not be read as achieved impact.

Saved per year across the cluster	
Fresh water	75 million litres
Salt	4,500 tonnes
Evaporator steam	14,106 MWh
CO2	6,376 tonnes

Environmental benefit

Less groundwater pulled out of a basin that is already stressed. Less coal burned for the same output. Less solid salt added to the 60,000 to 73,000 tonnes already stockpiled across Tirupur's treatment plants

with no settled disposal route.

Economic benefit

A lower cost base for a cluster that exports around Rs 46,000 crore a year and competes on price. The 2011 shutdown is living memory in Tirupur, so anything that strengthens environmental compliance also protects the jobs that depend on staying open.

The policy finding

My system can compute what price would make the bigger saving worth buying. It walks each price up or down, re-running the whole calculation at every step, and finds the point where a factory chasing profit picks the 47% plan on its own.

Price	Today	Changes at	Move needed
Low-salt dye premium	Rs 6.50/kg fabric	Rs 5.74/kg fabric	12% cheaper
Boiler steam	Rs 2.03/kWh heat	Rs 3.13/kWh heat	54% dearer
Fresh water	Rs 45/m3	Rs 251/m3	457% dearer

The steam row converts into a carbon price. Steam would need to rise Rs 1.09 per kWh of heat, and making that heat gives off 0.452 kg of CO₂, so the carbon price that produces the rise is **Rs 2,423 a tonne, about 21 euro**.

The EU carbon market was charging 82.40 euro a tonne on 5 October 2026, close to four times more. So this saving is not waiting for an impossible price. It is waiting for any price at all.

That is a finding a regulator can act on, and it came out of the engine rather than out of an opinion.

11. Conclusion

ZLD saved the Noyyal river and left the cluster with a coal bill nobody is managing. That bill is set by salt, not water, so most of the water-saving work in Tirupur is aimed at the wrong number. ChangeLoop shows the planner what their choice actually costs, at the moment they make it.

The prototype is built, running and open to inspection. The physics is checked against published plant measurements. The business case pays back in under three years on the conservative plan, and in weeks if a small price gap closes. The route to the cluster runs through 18 shared treatment plants whose own economics already point the same way.

What it does not have is a single measured number from a real factory, and I would rather say that plainly than let a reviewer discover it. That is exactly what the funding is for.

What I am asking for

- **Rs 23,00,000** for a 12-month pilot across two to three Tirupur dyeing units
- **An introduction to a shared treatment plant** willing to host the pilot
- **Technical review** of the model by someone who knows textile effluent

Next steps

I can demonstrate the live system in under ten minutes, including the parts that show its own limits. I would welcome a technical review of the coefficients before any pilot begins, and I am happy to be told a number is wrong. Every one of them is open at the link below.

<https://changeloop-water-intelligence.onrender.com>

12. Appendix

How the figures in this note can be checked

Every engine figure quoted here is produced by the live system and can be reproduced. The input numbers, with their sources and arithmetic, are at

<https://changeloop-water-intelligence.onrender.com/api/evidence>

A script in the repository re-checks every number in this document against the engine and fails if any has drifted.

Evidence position

Label	How many	What it means
Published	17	From CPCB, the Central Electricity Authority, Tamil Nadu tariff orders, state pollution board data, the EU reference document, steam tables
Derived	4	Worked out from those by arithmetic a test re-checks
Assumed	4	My assumptions, labelled as such. All four are commercial prices public sources do not settle
Measured	0	No instrument on a real factory has fed this system. A test enforces this

What this project has not done

- No deployment, no customer, no pilot, no measured result.
- The cluster figure is a projection, not achieved impact.
- The financial projections in section 9 are plans with stated assumptions.
- The shared-evaporator solver checks every combination, which will not scale to a treatment plant with hundreds of members.
- The 60,000 mg/L limit is one modelled figure inside a published range of 15,000 to 80,000. Whether it holds in real running conditions is unknown.
- The data centre and cooling tower idea is an argument from structure, not a result.
- There is no trained model in this system and I do not call it AI-powered. The decision record is a hash chain, not a blockchain.

Sources

Figure	Where it comes from
Tirupur exports of about Rs 46,000 crore in FY26	Tiruppur Exporters Association, reported March 2026
300 dyeing units on 18 CETPs, 60 with their own plants, 120 MLD recycled, 94% recovery	Reported cluster data, 2026. Other reports put the unit count nearer 450

Figure	Where it comes from
Waste salt backlog of 60,000 to 73,000 tonnes across the 18 CETPs	Dyers Association of Tiruppur, reported by Ecotextile News, 2022, and later reporting on accumulated salt at the CETPs
EU carbon border mechanism covers iron and steel, cement, aluminium, fertilisers, electricity and hydrogen, not textiles	European Commission scope; textiles named as a candidate for a later phase
2011 closure of around 700 units	Madras High Court order, widely reported
Water strength of 18,340 mg/L, leftover at 20-30%, evaporator steam use	Central Pollution Control Board assessment of textile dyeing units and ZLD at Tirupur
Grid emissions, 0.705 kg CO ₂ per kWh	CO ₂ Baseline Database for the Indian Power Sector, Central Electricity Authority
Carbon price of 82.40 euro a tonne, 5 October 2026	European carbon prices, S&P Global Commodity Insights
Water and steam tariffs	Tamil Nadu Electricity Regulatory Commission tariff orders and state pollution board plant data
Latent heat, 0.62694 kWh/kg	Steam tables: 2,257 kJ/kg at 100 C and one atmosphere, divided by 3,600
Evaporator energy band, 150-250 kWh/m ³	EU Best Available Techniques reference document for textiles. Used as a check, never as an input